MACHINE LEARNING & CYBER SECURITY PROJECT

Phishing Website Detection

Project Overview

This project trains a **Random Forest** machine-learning model to detect phishing website records using the UCI Phishing Websites dataset. The primary goal is to demonstrate how machine learning can identify patterns associated with malicious and phishing websites to support basic cybersecurity awareness.

DATASET SPECIFICATION

Attribute	Details
Name	Phishing Websites
Source	UCI Machine Learning Repository
Dataset ID	327
Total Records	11,055
Feature Count	30 Features

TARGET VARIABLES

The model classifies websites using the **Result** variable:

- `-1` **Phishing Website:** Malicious link identified.
- `1` **Legitimate Website:** Safe link identified.

Note: The dataset downloads automatically via the `ucimlrepo` Python library.

Technologies Used

Python 3.10+ | pandas | scikit-learn | Matplotlib | ucimlrepo | joblib

Project Deliverable Files

- `RamanKumar_PhishingWebsiteDetection.py` — Main Python execution script
- `requirements.txt` — Python dependency management file
- `RamanKumar_ProjectReport.docx` — Detailed project documentation
- `README.md` — Technical setup and project overview

Setup and Installation

1. Install **Python 3.10** or a newer version on your system.
2. Open a terminal inside your working project directory.
3. Create and activate a isolated virtual environment:

```
python -m venv venv
```

Windows: `venv\Scripts\activate` | **macOS/Linux:** `source venv/bin/activate`

4. Install all required project dependencies:

```
pip install -r requirements.txt
```

5. Execute the machine learning pipeline script:

```
python RamanKumar_PhishingWebsiteDetection.py
```

** An active internet connection is only required during the initial dataset download.*

Program Execution Flow & Outputs

EXECUTION STEPS

1. Downloads dataset from UCI Repository.
2. Separates input features and target labels.
3. Splits records into training & testing sets.
4. Trains Random Forest classifier model.
5. Generates accuracy & evaluation report.
6. Saves model file & visual charts.

EXPECTED OUTPUTS

- Dataset summary information
- Model Accuracy Score
- Detailed Classification Report
- Confusion Matrix Visual Chart
- Feature Importance Chart
- `phishing_random_forest_model.joblib`

Important Security Notice

This project is created strictly for learning and demonstration purposes. Model predictions can produce errors and must not be used as the sole deciding factor for website safety. Always manually verify website URLs and rely on established security tools.

Reference

Mohammad, R., McCluskey, T. L., and Thabtah, F.